

Sri Lanka Institute of Information Technology

IT3011 – TPSM

Final Report



Group Name: The Learning Loop

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1. Introduction

This report investigates the statement: "**Students who reflect on their learning demonstrate deeper understanding.**" The goal is to find out whether students who show reflective learning behaviors actually perform better academically, and to test this claim using real student data and statistical methods.

To do this, we work with one dataset and two research papers. The dataset - student_performance.csv contains academic and behavioral records of students and is used for all statistical modelling in this report. The two research papers, Dahlgren (2020) and Lew & Schmidt (2011), are used for justification purposes - meaning we refer to their findings to support or contextualize the results we get from our own analysis, rather than modelling their data directly.

In this study, reflection is not something we can measure directly from the dataset. Instead, we identify five variables that represent reflective learning behaviors, which we call **reflection proxies**. These are Study Hours, Attendance, Discussions, Motivation, and Assignment Completion. Students who study consistently, attend classes, participate in discussions, stay motivated, and complete their assignments are considered to be engaging in reflective learning behaviors.

To measure deeper understanding, we use two **understanding proxies** - Exam Score and Final Grade. Exam Score represents how well a student performed on the exam, and Final Grade represents the student's overall academic outcome across the semester.

The analysis is structured in three stages. First, **descriptive analysis** is used to summarize and visualize the data. Second, **inferential analysis** is used to statistically test whether the reflection proxies have a significant relationship with the understanding proxies. Third, **predictive analysis** builds a model to see how well the reflection proxies can predict student performance.

The hypotheses guiding this analysis are:

- **H₀ (Null Hypothesis):** Reflection does not significantly affect deeper understanding.
- **H₁ (Alternative Hypothesis):** Reflection significantly affects deeper understanding.

A significance level of $\alpha = 0.05$ is used for all hypothesis tests throughout this report.

2. Data Cleaning

The raw dataset, `student_performance.csv`, contained 14,003 rows and 16 columns covering student academic and behavioral characteristics. The columns included study hours, attendance, resource usage, extracurricular participation, motivation, internet access, gender, age, learning style, online course participation, discussions, assignment completion, exam scores, educational technology usage, stress level, and final grade. Before any analysis could begin, the dataset was cleaned to ensure the data was accurate, consistent, and suitable for modelling. The following steps were carried out in R:

1. **Checking dimensions and structure:** The dataset was loaded into R and confirmed to have 16 columns. The structure and summary of all columns were reviewed to get a general understanding of the data and to spot anything unusual.
2. **Checking for missing values:** No missing values (NA entries) were found in any column. Therefore, no imputation was required.
3. **Checking for impossible values:** The Attendance column was checked for any values above 100, as attendance cannot logically exceed 100%. No such values were found, confirming the data was clean on this front.
4. **Removing duplicate rows:** Duplicate rows were checked using R's `duplicated()` function. Duplicates were found and removed using `distinct()`. After removing duplicates from the full 16-column dataset, a subset of 7 columns was created. A second round of duplication was applied after subsetting, because dropping 9 columns caused previously unique rows to become identical. After both rounds of duplication, the dataset was reduced to **7,813 rows**.
5. **Converting FinalGrade to a factor:** The FinalGrade column contained values 0, 1, 2, and 3. Although stored as numbers, these represent grade categories rather than quantities. It was converted to a factor in R so that it is treated correctly in visualizations and statistical models.
6. **Creating the working subset:** Seven columns were selected for analysis: StudyHours, Attendance, Discussions, Motivation, AssignmentCompletion, ExamScore, and FinalGrade. These represent the five reflection proxies and two understanding proxies identified for this study.
7. **Saving the cleaned file:** The cleaned subset was saved as `cleaned_student_data.csv`, which was used as the starting point for all subsequent analysis.

Summary of changes made:

Table 1: Summary of Data Cleaning Actions Taken

Check	Finding	Action Taken
Missing values	None found	No action needed
Impossible values	None found	No action needed
Duplicate rows	Found and removed (2 rounds)	Removed using <code>distinct()</code>
FinalGrade column	Stored as integer	Converted to factor with 4 levels
Dataset size	14,003 rows, 16 columns → 7,813 rows, 7 columns	Subset and deduplicated

3. Data Description

The cleaned dataset used for this analysis contains 7,813 student records and 7 variables - five reflection proxies (StudyHours, Attendance, Discussions, Motivation, AssignmentCompletion) and two understanding proxies (ExamScore and FinalGrade).

The table below provides a brief description of each variable:

Table 2: Description of Variables in the Cleaned Dataset

Variable	Type	Description
StudyHours	Numeric	Number of hours the student spends studying
Attendance	Numeric	Attendance percentage (0–100)
Discussions	Binary	Whether the student participated in discussions (0 = No, 1 = Yes)
Motivation	Ordinal	Motivation level of the student (0, 1, or 2)
AssignmentCompletion	Numeric	Percentage of assignments completed (0–100)
ExamScore	Numeric	Score obtained in the exam (0–100)
FinalGrade	Factor	Overall final grade category (0, 1, 2, or 3)

FinalGrade is treated as a categorical variable throughout this report. Grade 0 represents the highest performing students and Grade 3 represents the lowest, based on the pattern observed in the data.

4. Descriptive Analysis

Before running any statistical tests, it is important to first understand what the data looks like. Descriptive analysis does exactly this - it summarizes each variable using key statistics and visualizes the distributions and relationships between variables. This helps us spot patterns, understand the spread of the data, and form early impressions about whether any relationships exist between the reflection proxies and understanding proxies. At this stage we are not testing anything formally. We are simply exploring the data.

Table 3: Summary Statistics for Numeric Variables

	Column	Mean	Median	SD	Minimum	Maximum
<i>StudyHours</i>	StudyHours	20.03	20	6.70	5	44
<i>Attendance</i>	Attendance	80.11	80	11.59	60	100
<i>Discussions</i>	Discussions	0.61	1	0.49	0	1
<i>Motivation</i>	Motivation	0.93	1	0.75	0	2
<i>AssignmentCompletion</i>	AssignmentCompletion	74.55	74	14.67	50	100
<i>ExamScore</i>	ExamScore	70.37	70	17.66	40	100

4.1 Chart 1 - Histogram of Exam Score

Exam scores are spread fairly evenly across the range of 40 to 100, with each 5-point scoring band containing roughly 600 to 750 students. There is no strong peak in the middle. The distribution is approximately uniform, meaning students are spread across all performance levels rather than clustering around any particular score. The only slight exception is the first bar (40–45) which is marginally higher than the rest, but the overall pattern remains largely flat.

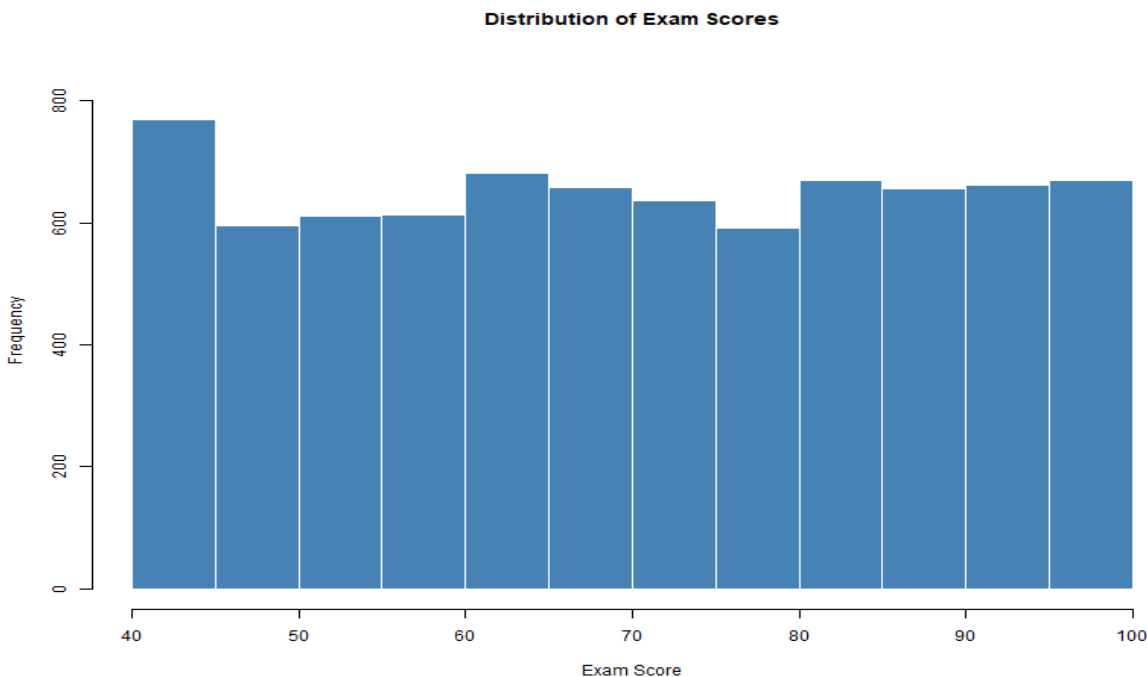


Figure 1: Distribution of Exam Scores

4.2 Chart 2 - Bar Chart of Mean ExamScore by Discussions

Students who did not participate in discussions recorded a slightly higher average exam score (70.96) compared to those who did participate (69.99). The difference is minimal. Less than 1 point suggesting that discussion participation alone has no meaningful impact on exam performance in this dataset.

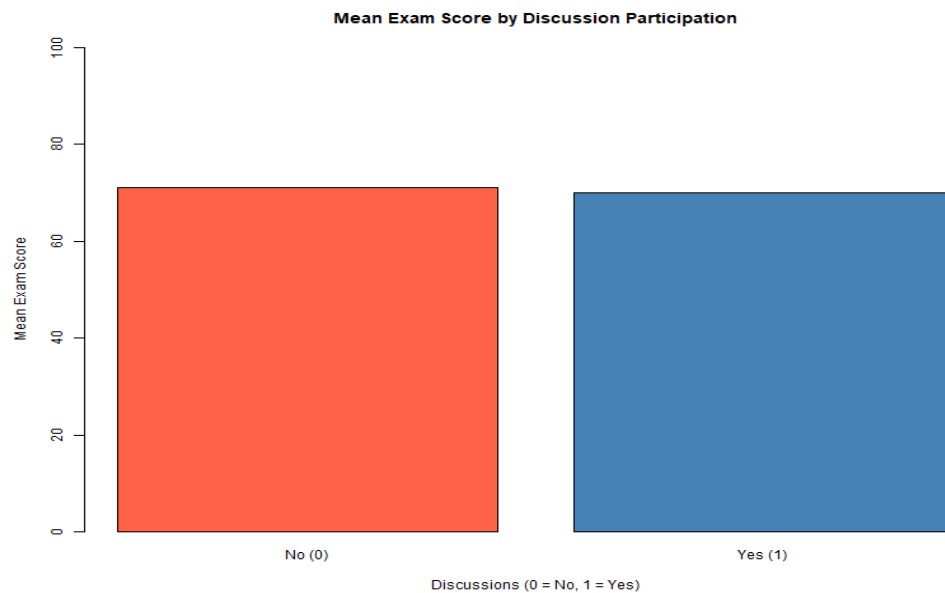


Figure 2: Mean Exam Score by Discussion Participation

4.3 Chart 3 - Box Plot of ExamScore by FinalGrade

There is a clear and consistent pattern. Students with Grade 0 scored highest (median 92, range 85–100), followed by Grade 1 (median 77), Grade 2 (median 62), and Grade 3 the lowest (median 47, range 40–54). The boxes across all four grade groups are similarly narrow, indicating that exam scores within each grade category are tightly clustered with little variation.

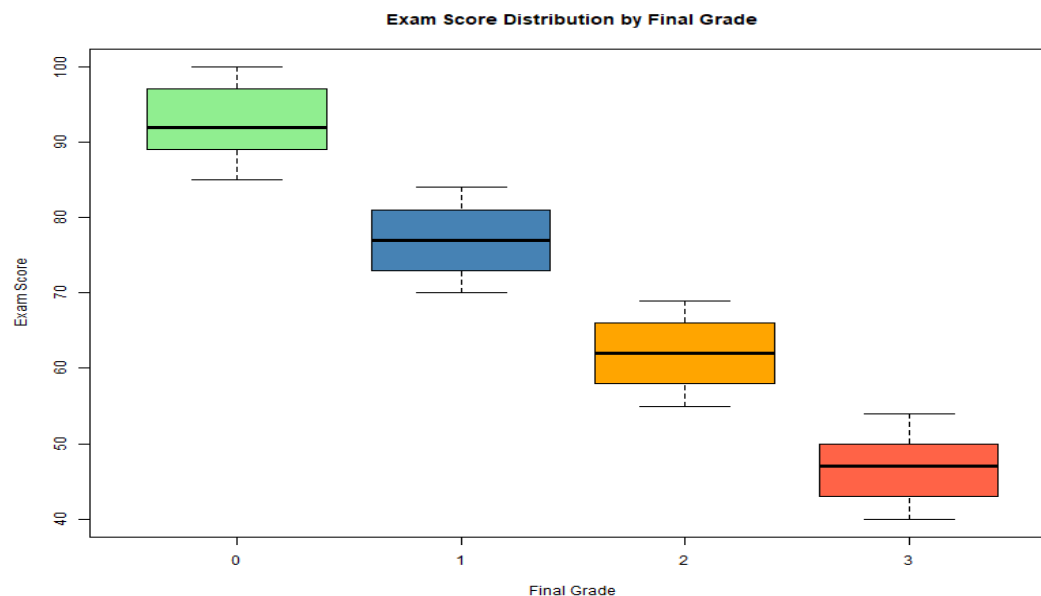


Figure 3: Exam Score Distribution by Final Grade

4.4 Chart 4 - Correlation Table

Nearly all correlations between the six numeric variables are extremely close to zero, indicating that none of the variables have a meaningful linear relationship with each other in this dataset. The strongest correlation observed is between AssignmentCompletion and ExamScore (0.018) and between Attendance and StudyHours (0.034), but both are so weak they are practically negligible.

	StudyHours	Attendance	Discussions	Motivation	AssignmentCompletion	ExamScore
StudyHours	1	0.034	0.021	-0.004	0.008	0.002
Attendance	0.034	1	-0.009	0.003	0.001	-0.013
Discussions	0.021	-0.009	1	-0.013	0.007	-0.027
Motivation	-0.004	0.003	-0.013	1	-0.001	-0.005
AssignmentCompletion	0.008	0.001	0.007	-0.001	1	0.018
ExamScore	0.002	-0.013	-0.027	-0.005	0.018	1

Figure 4: Correlation Matrix of Numeric Variables - Table View

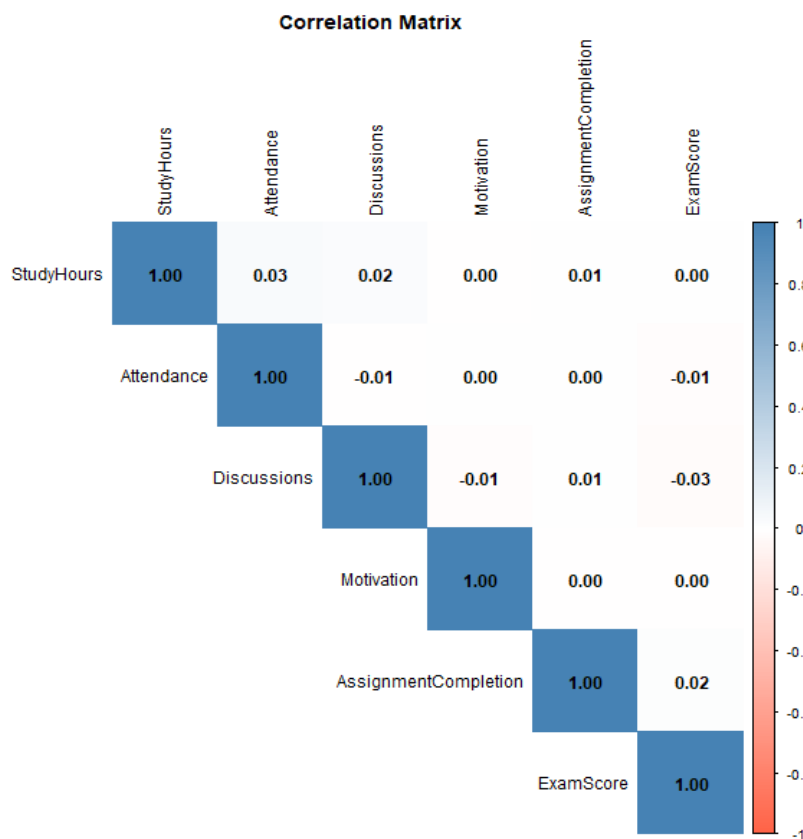


Figure 5: Correlation Matrix of Numeric Variables - Heatmap

4.5 Closing Summary

Overall, the descriptive analysis gives a first look at the data without drawing any firm conclusions. The box plot is the most informative chart at this stage, clearly showing that students in higher final grade categories consistently score higher on exams with very little overlap between groups. These early observations set the stage for the inferential analysis, where we will formally test whether any of these patterns are statistically significant.

5. Inferential Analysis

Inferential analysis goes beyond simply describing the data. It allows us to test whether the patterns observed are statistically meaningful or whether they could have occurred by chance. In this section, we apply three statistical tests to examine the relationship between the five reflection proxies and the two understanding proxies. All tests are conducted at a significance level of $\alpha = 0.05$, meaning a result is considered statistically significant only if the probability of it occurring by chance is less than 5%.

5.1 Pearson Correlation - Reflection proxies vs ExamScore

Pearson correlation measures the strength and direction of the linear relationship between two continuous numeric variables. It produces a value called r (correlation value), ranging from -1 to +1, where values close to 0 indicate no relationship, values close to +1 indicate a strong positive relationship, and values close to -1 indicate a strong negative relationship.

ExamScore was selected as the first understanding proxy for this test because it is a continuous numeric variable, which satisfies the requirement of Pearson correlation. All five reflection proxies are also measured on a numeric scale, making Pearson the appropriate test for examining their individual relationships with ExamScore.

- p-value – if below 0.05, the relationship is statistically significant.
- **Reject H_0** if p-value < alpha (0.05)

The hypotheses for each correlation test are:

- H_0 : There is no significant linear relationship between the reflection proxy and ExamScore
- H_1 : There is a significant linear relationship between the reflection proxy and ExamScore

Table 4: Pearson Correlation Results - Reflection Proxies vs ExamScore

Reflection Proxy	r value	t-statistic	p-value	Significant?	What it means	Decision
StudyHours	0.0015	0.13312	0.8941	No	Practically zero relationship	Do not reject H_0
Attendance	-0.0127	-1.1237	0.2612	No	No meaningful relationship	Do not reject H_0
Discussions	-0.0268	-2.3673	0.0179	Yes	Weak but statistically significant	Reject H_0
Motivation	-0.0055	-0.48524	0.6275	No	No relationship	Do not reject H_0
Assignment Completion	0.0183	1.6143	0.1065	No	No significant relationship	Do not reject H_0

The results show that four out of five reflection proxies produced p-values greater than 0.05, leading us to fail to reject H_0 for these variables. This indicates no statistically significant linear relationship between these proxies and exam performance. Discussions was the only proxy that produced a significant result ($p = 0.018$), however the correlation coefficient of $r = -0.027$ is extremely close to zero, indicating the relationship, while statistically detectable, is negligible in practical terms. This finding is consistent with Lew & Schmidt (2011), who similarly reported weak correlations between reflection-related behaviors and academic test grades across a large student sample.

5.2 Spearman Correlation - Reflection proxies vs FinalGrade

While Pearson correlation requires both variables to be continuous and numeric, FinalGrade in this dataset is an ordinal categorical variable with four levels representing grade categories (0, 1, 2, 3). Applying Pearson correlation to a categorical variable would produce mathematically invalid results. For this reason, Spearman rank correlation was used instead, as it is designed for situations where one or both variables are ordinal. Spearman works by ranking the values rather than using raw numbers, making it suitable for this type of data. This is also why FinalGrade was converted to a factor during data cleaning, to correctly identify it as a categorical variable rather than a quantity.

- rho value – close to 0 = weak, close to ± 1 = strong
- p-value – if below 0.05, the relationship is statistically significant
- **Reject H_0** if p-value < alpha (0.05)

The hypotheses for each test are:

- H_0 : There is no significant relationship between the reflection proxy and FinalGrade
- H_1 : There is a significant relationship between the reflection proxy and FinalGrade

Table 5: Spearman Correlation Results - Reflection Proxies vs FinalGrade

Reflection Proxy	rho value	p-value	Significant?	What it means	Decision
StudyHours	-0.0051	0.649	No	No relationship with FinalGrade	Do not reject H_0
Attendance	0.0105	0.355	No	No relationship	Do not reject H_0
Discussions	0.0343	0.002	Yes	Weak but significant	Reject H_0
Motivation	0.0056	0.619	No	No relationship	Do not reject H_0
AssignmentCompletion	-0.0201	0.075	No	Borderline - just above 0.05	Do not reject H_0

The results mirror those of the Pearson correlation. Four proxies (StudyHours, Attendance, Motivation, and AssignmentCompletion) show no statistically significant relationship with FinalGrade, as all p-values exceed 0.05. Discussions again produced a significant result ($p = 0.002$), with a Spearman rho of 0.034. While this confirms a real relationship exists, the effect size remains very small.

AssignmentCompletion approached significance with a p-value of 0.075 but did not cross the threshold. The warning message produced by R stating "*Cannot compute exact p-value with ties*" is a known limitation when many students share the same FinalGrade category. The approximate p-values reported remain valid for interpretation purposes.

5.3 Independent Samples T-Test

The independent samples t-test compares the mean ExamScore of two separate, independent groups to determine whether the difference between them is statistically significant. Among the five reflection proxies, Discussions is the only variable that naturally divides students into exactly two groups as those who participated (coded as 1) and those who did not (coded as 0). The remaining proxies such as StudyHours and Attendance are continuous variables that cannot be split into two clean groups, making the t-test unsuitable for them. This is why the t-test is applied specifically to Discussions as the reflection proxy, with ExamScore as the outcome.

The hypotheses are:

- H_0 : There is no significant difference in mean ExamScore between students who participated in discussions and those who did not
- H_1 : There is a significant difference in mean ExamScore between the two groups

Table 6: Independent Samples T-Test Results - ExamScore by Discussion Group

Group	Mean ExamScore	t-statistic	df (degrees of freedom)	p-value	Decision
Discussions = 0 (Did not participate)	70.96	2.3647	6474.5	0.018	Reject H_0
Discussions = 1 (Participated)	69.99	2.3647	6474.5	0.018	Reject H_0

The test produced a t-statistic of 2.365 with a p-value of 0.018, which falls below the significance level of 0.05. Therefore, we reject H_0 and conclude that a statistically significant difference exists between the two groups. However, the practical significance of this result must be interpreted carefully. Students who did not participate in discussions recorded a mean ExamScore of 70.96, while those who did participate recorded 69.99 - a difference of less than one mark. While the large sample size of over 7,800 students makes even tiny differences detectable, a difference of this magnitude has no meaningful real-world implication. Dahlgren (2020) noted similarly that reflection-based strategies do not uniformly benefit all students, which may explain why discussion participation does not translate into a meaningful score advantage.

5.4 One-Way ANOVA + Post-Hoc Analysis

One-way Analysis of Variance (ANOVA) tests whether the mean of a continuous variable differs significantly across three or more groups. It is used here to examine whether students in different FinalGrade categories (0, 1, 2, and 3) score significantly differently on their exams. Unlike the t-test which handles two groups, ANOVA extends this comparison to multiple groups simultaneously while controlling the overall error rate. FinalGrade is used as the grouping variable here because it represents the overall academic outcome of the student, which is influenced by multiple factors beyond just exam performance, making it meaningful to test whether it separates students by exam score in a statistically significant way.

The hypotheses are:

- H_0 : The mean ExamScore is equal across all four FinalGrade groups
- H_1 : At least one FinalGrade group has a significantly different mean ExamScore

The ANOVA produced an F-statistic of 38,843 with a p-value below 0.001, providing overwhelming evidence to reject H_0 . This confirms that mean ExamScore differs significantly across four grade groups.

A post-hoc Tukey HSD test was conducted to identify which specific grade group pairs differed from each other. The results confirmed that every pair of grade groups was significantly different, with each adjacent grade level separated by approximately 15 exam marks.

Table 7: One-Way ANOVA Results - ExamScore by FinalGrade

Source of Variation	df	Sum Sq (SS)	Mean Sq (MS)	F value	p-value
Between Groups (FinalGrade)	3	2,283,555	761,185	38,843	< 0.001
Within Groups (Residuals)	7,809	153,028	20		
Total	7812	2,436,583			

Since $p\text{-value} < 0.05$, we **reject H_0**

5.5 Overall Interpretation

Taken together, the inferential results provide only limited and partial support for H_1 . Across ten correlation tests covering all five reflection proxies against both understanding proxies, only Discussions produced statistically significant results and even then, the effect sizes were negligible ($r = -0.027$, $\rho = 0.034$). The t-test confirmed a significant difference between discussion groups but with a practically meaningless gap of under one mark. The ANOVA result was highly significant, confirming that the two understanding proxies are genuinely differentiated by student performance levels.

These findings align closely with the broader literature. Lew & Schmidt (2011) reported that self-reflection showed weak to moderate correlations with academic performance across 690 students, concluding that while reflection does support learning, its effect on measurable academic outcomes is subtle. Dahlgren (2020) similarly noted that reflective strategies are not equally effective for all students. The behavioral proxies used in this study such as study hours and attendance are indirect measures of reflection rather than direct ones, which likely contributes to the weak relationships observed. This limitation is acknowledged and discussed further in Section 7.

6. Predictive Analysis

6.1 Overview

While inferential analysis tells us whether relationships between variables are statistically significant, predictive analysis goes one step further. It uses those variables to build a model that can actually predict outcomes for students. In this section, two regression models are developed using the five reflection proxies as predictor variables. The first model predicts ExamScore as a continuous outcome, and the second predicts whether a student is likely to pass or fail based on their FinalGrade category. Together, these models allow us to assess whether reflection-related behaviors have enough predictive power to meaningfully forecast student understanding.

6.2 Model 1 - Linear Regression (Predicting ExamScore)

Linear regression models the relationship between a continuous outcome variable and one or more predictor variables by fitting a straight line through the data. It produces a predicted ExamScore for each student based on their values across the five reflection proxies. This model is appropriate here because ExamScore is a continuous numeric variable ranging from 40 to 100.

All five reflection proxies are included as predictors, as the purpose is to assess their combined ability to explain variation in exam performance.

Table 8: Linear Regression Coefficients - Predicting ExamScore

Reflection Proxy	Coefficient	Standard error	t-value	p-value
Intercept	70.90275	1.841229	38.508	< 0.001
StudyHours	0.006166	0.029844	0.207	0.8363
Attendance	-0.019857	0.017245	-1.151	0.2496
Discussions	-0.982277	0.409593	-2.398	0.0165
Motivation	-0.136298	0.266393	-0.512	0.6089
AssignmentCompletion	0.022199	0.013620	1.630	0.1032

The R-squared value of 0.0013 indicates that the five reflection proxies combined explain less than 0.2% of the variation in ExamScore. The overall model is not statistically significant (F-statistic p-value = 0.079, which is above 0.05), meaning the model as a whole fails to predict exam performance better than chance. Among individual predictors, only Discussions showed a significant coefficient ($p = 0.017$), consistent with the inferential analysis findings. This extremely low explanatory power confirms that study habits and behavioral engagement indicators alone are insufficient predictors of exam performance.

- Multiple R squared: 0.001264
- Adjusted R-squared: 0.0006246
- Residual Std Error: 17.6552

Given that the correlations from the inferential analysis were all very weak, the R-squared value is expected to be very low - meaning the five reflection proxies together do not explain much of the variation in ExamScore. This is an important and honest finding. It suggests that exam performance is driven by factors beyond the behavioral indicators measured in this dataset, which aligns with Lew & Schmidt (2011) who noted that the link between reflective behaviors and measurable academic outcomes is subtle and limited.

6.3 Model 2 - Logistic Regression (Predicting Pass/Fail)

Unlike linear regression which predicts a number, logistic regression predicts the probability of belonging to one of two categories. Here, FinalGrade is converted into a binary Pass/Fail variable. Students with FinalGrade 0 or 1 are classified as Pass (0) and students with FinalGrade 2 or 3 are classified as Fail (1). Logistic regression is the correct model for this type of binary outcome because it keeps predicted probabilities between 0 and 1, which linear regression cannot guarantee.

The same five reflection proxies are used as predictors to maintain consistency with Model 1 and to examine whether these variables can classify students into the correct pass or fail category.

Table 9: Logistic Regression Coefficients - Predicting Pass/Fail

Reflection Proxy	Coefficient	Standard error	z-value	p-value
StudyHours	-0.0011001	0.0033835	-0.325	0.7451
Attendance	0.0002791	0.0019552	0.143	0.8865
Discussions	0.1117723	0.0464544	2.406	0.0161
Motivation	0.0231236	0.0302020	0.766	0.4439
AssignmentCompletion	-0.0022300	0.0015444	-1.444	0.1488

The model assigns each student a predicted probability of failing. If this probability exceeds 0.5, the student is classified as predicted to fail; otherwise, they are predicted to pass. The confusion matrix below shows how well the model's predictions match the actual outcomes.

Table 10: Confusion Matrix - Logistic Regression Predictions vs Actual Outcomes

	Actual 0	Actual 1
Predicted 0	2875	2690
Predicted 1	1123	1125

The confusion matrix compares what the model predicted against what actually happened for all 7,813 students. The model correctly predicted 2875 students as Pass and 1125 students as Fail, giving a total of 4,000 correct predictions. The remaining 3813 students were misclassified. 2690 students who actually failed were incorrectly predicted to pass, and 1123 students who actually passed were incorrectly predicted to fail. Model accuracy is calculated as the number of correct predictions divided by the total number of students: $4000 \div 7813 = 51.2\%$. This is only marginally better than random guessing, which would give 50%, confirming that the reflection proxies have very limited ability to correctly classify student outcomes.

Table 11: Odds Ratios - Logistic Regression Predictors

Variable	Odds Ratio	What it means
StudyHours	0.9989	Almost no effect
Attendance	1.0003	Almost no effect
Discussions	1.1183	Participating in discussions increases odds of failing by about 12%
Motivation	1.0234	Almost no effect
AssignmentCompletion	0.9978	Almost no effect

An odds ratio of exactly 1.0 means the variable has no effect on the probability of failing. A value above 1.0 means higher values of that variable increase the probability of failing, while a value below 1.0 means it decreases it. As shown in the table, all five reflection proxies produce odds ratios extremely close to 1.0, confirming that none of them meaningfully influence whether a student passes or fails. Discussions has the highest odds ratio at 1.1183, meaning that participating in discussions is associated with approximately a 12% increase in the odds of failing. This is consistent with the t-test finding in the inferential analysis, where discussion participants scored very slightly lower on exams than non-participants. However, given that all other proxies show virtually no effect and the overall model accuracy is only 51.2%, this result should not be interpreted as a meaningful or actionable finding.

7. Interpretation & Justification

This section brings together the findings from all three stages of analysis - descriptive, inferential, and predictive - and evaluates them against the original statement: *"Students who reflect on their learning demonstrate deeper understanding."* The results are interpreted in the context of existing literature to assess whether the data supports, partially supports, or contradicts this statement.

7.1 What the Descriptive Analysis Showed

The descriptive analysis showed exam scores spread evenly across all students, with the box plot revealing a clear separation between grade groups, and the bar chart suggesting almost no difference in exam scores between discussion participants and non-participants.

7.2 What the Inferential Analysis Showed

The inferential analysis found that only Discussions showed statistically significant relationships across all tests, but with effect sizes too small to be practically meaningful. All other four reflection proxies showed no significant relationship with either understanding proxy.

7.3 What the Predictive Analysis Showed

Both regression models confirmed this pattern. The linear regression explained only 0.13% of variation in ExamScore, and the logistic regression achieved 51.2% accuracy, barely above random chance.

7.4 Justification Using Literature

These findings are not unusual or unexpected. Lew & Schmidt (2011) conducted a large-scale study with 690 students and found similarly weak correlations between reflective behaviors and academic grades, concluding that self-reflection improves learning *to a limited extent* and that its impact on measurable academic outcomes is subtle. Dahlgren (2020) found that while most students felt reflective practices helped them self-regulate, the strategy was not equally effective for all learners, and motivational factors played a significant role in whether reflection translated into academic improvement.

A key reason for the weak results in this study is that the proxy variables used such as study hours, attendance, and assignment completion are indirect and behavioral measures of reflection, not direct measures. They capture what students do, not how deeply they think about their learning. Lew & Schmidt (2011) used automated text analysis of actual student reflection journals, which is a far more direct measurement. The gap between behavioral proxies and genuine reflective thinking likely explains the negligible relationships observed throughout this analysis.

7.5 Overall Position on H_0 and H_1

Based on the combined evidence, H_0 cannot be fully rejected. The data provides only weak and inconsistent support for H_1 . Only one reflection proxy out of five showed statistically significant results across tests, and even that proxy produced effect sizes too small to support a meaningful conclusion. The statement "*students who reflect on their learning demonstrate deeper understanding*" may hold true in principle, as supported by existing literature, but the behavioral proxy variables used in this dataset are insufficient to statistically demonstrate that relationship.

8. Conclusion

This study set out to examine whether student reflection, measured through five behavioral proxies, is associated with deeper academic understanding, measured through exam scores and final grade outcomes. Three stages of statistical analysis - descriptive, inferential, and predictive were applied to a cleaned dataset of 7,813 students.

The overall conclusion is that the data provides **limited and partial support** for the hypothesis that reflection significantly affects deeper understanding. This finding is consistent with the broader academic literature - Lew & Schmidt (2011) and Dahlgren (2020) both found that the relationship between reflective behavior and measurable academic outcomes is subtle and not easily captured through standardized measures.

The primary limitation of this study is the use of indirect behavioral proxies rather than direct reflection measures. Future research could improve on this by incorporating qualitative reflection data, self-assessment scores, or structured reflection journal analysis, as used in Lew & Schmidt (2011), which may reveal stronger and more meaningful relationships between reflective practice and academic understanding.

9. References

Dahlgren, J. (2020). *Reflection's impact on student learning* [Master's project, Minnesota State University Moorhead]. RED: A Repository of Digital Collections.

Lew, M. D. N., & Schmidt, H. G. (2011). Self-reflection and academic performance: Is there a relationship? *Advances in Health Sciences Education*, 16(5), 529–545.

10. Appendix

Appendix A – Data Cleaning Script

```
library(dplyr)

# Load CSV file
data <- read.csv("C:\\Users\\yazee\\OneDrive - Sri Lanka Institute of Information Technology
(1)\\Y3S1\\IT3011 - TSPM\\TSPM - Assignment\\student_performance.csv")

# Check dimensions
dim(data)
str(data)
summary(data)

# Check missing values
colSums(is.na(data))

# Check impossible values
data[data$Attendance > 100, ]

# Check duplicates
sum(duplicated(data))

# Remove duplicates
data <- distinct(data)

# Convert FinalGrade to factor
data$FinalGrade <- as.factor(data$FinalGrade)
class(data$FinalGrade)
levels(data$FinalGrade)

# Create 7-column subset
clean_data <- data[, c("StudyHours", "Attendance", "Discussions",
                      "Motivation", "AssignmentCompletion",
                      "ExamScore", "FinalGrade")]

# Remove duplicates again after subsetting
clean_data <- distinct(clean_data)

# Verify
dim(clean_data)
colnames(clean_data)

# Save
write.csv(clean_data, "cleaned_student_data.csv", row.names = FALSE)

# Check current working directory
getwd()
```

Appendix B - Descriptive Analysis Script

```
library(dplyr)
library(corrplot)

# Load cleaned data
clean_data <- read.csv("C:\\Users\\yazee\\OneDrive - Sri Lanka Institute of Information
Technology (1)\\Y3S1\\IT3011 - TPSM\\TPSM - Assignment\\cleaned_student_data.csv")
clean_data$FinalGrade <- as.factor(clean_data$FinalGrade)

numeric_cols <- clean_data[, c("StudyHours", "Attendance", "Discussions",
                               "Motivation", "AssignmentCompletion", "ExamScore")]

# Summary Statistics Table
summary_stats <- data.frame(
  Column = names(numeric_cols),
  Mean = round(sapply(numeric_cols, mean), 2),
  Median = round(sapply(numeric_cols, median), 2),
  SD = round(sapply(numeric_cols, sd), 2),
  Minimum = round(sapply(numeric_cols, min), 2),
  Maximum = round(sapply(numeric_cols, max), 2)
)
print(summary_stats)

# Correlation Matrix
cor_matrix <- round(cor(numeric_cols), 3)
print(cor_matrix)

# Chart 1 – Histogram of ExamScore
hist(clean_data$ExamScore,
  main = "Distribution of Exam Scores",
  xlab = "Exam Score",
  ylab = "Frequency",
  col = "steelblue",
  border = "white",
  breaks = 6,
  ylim = c(0, max(hist(clean_data$ExamScore,
                        breaks = 12, plot = FALSE)$counts) * 1.1))

# Chart 2 – Bar Chart of Mean ExamScore by Discussions
mean_by_disc <- tapply(clean_data$ExamScore, clean_data$Discussions, mean)
barplot(mean_by_disc,
  main = "Mean Exam Score by Discussion Participation",
  xlab = "Discussions (0 = No, 1 = Yes)",
  ylab = "Mean Exam Score",
  col = c("tomato", "steelblue"),
  ylim = c(0, 100),
  names.arg = c("No (0)", "Yes (1)"))
```

```

# Chart 3 – Box Plot of ExamScore by FinalGrade
boxplot(ExamScore ~ FinalGrade,
        data = clean_data,
        main = "Exam Score Distribution by Final Grade",
        xlab = "Final Grade",
        ylab = "Exam Score",
        col = c("lightgreen", "steelblue", "orange", "tomato"))

# Chart 4 – Correlation Heatmap
corrplot(cor_matrix,
         method = "color",
         type = "upper",
         addCoef.col = "black",
         tl.col = "black",
         title = "Correlation Matrix",
         mar = c(0, 0, 1, 0),
         col = colorRampPalette(c("tomato", "white", "steelblue"))(200),
         cl.lim = c(-1, 1))

```

Appendix C - Inferential Analysis Script

```

# Load cleaned data
df <- read.csv("C:\\Users\\yazee\\OneDrive - Sri Lanka Institute of Information Technology
(1)\\Y3S1\\IT3011 - TSPM\\TSPM - Assignment\\cleaned_student_data.csv")
df$FinalGrade <- as.factor(df$FinalGrade)

# run pearson correlation between all 5 reflection proxies and examscore
cor.test(df$StudyHours, df$ExamScore, method = "pearson")
cor.test(df$Attendance, df$ExamScore, method = "pearson")
cor.test(df$Discussions, df$ExamScore, method = "pearson")
cor.test(df$Motivation, df$ExamScore, method = "pearson")
cor.test(df$AssignmentCompletion, df$ExamScore, method = "pearson")

# run spearman correlation between all 5 reflection proxies and finalgrade
cor.test(df$StudyHours, as.numeric(df$FinalGrade), method = "spearman")
cor.test(df$Attendance, as.numeric(df$FinalGrade), method = "spearman")
cor.test(df$Discussions, as.numeric(df$FinalGrade), method = "spearman")
cor.test(df$Motivation, as.numeric(df$FinalGrade), method = "spearman")
cor.test(df$AssignmentCompletion, as.numeric(df$FinalGrade), method = "spearman")

# Independent Samples T-test
t.test(ExamScore ~ Discussions, data = df)

model <- aov(ExamScore ~ FinalGrade, data = df)
summary(model)

# Post-hoc test – tells which groups differ from each other
TukeyHSD(model)

```

Appendix D - Predictive Analysis Script

```
# Load cleaned data
df <- read.csv("C:\\Users\\yazee\\OneDrive - Sri Lanka Institute of Information Technology
(1)\\Y3S1\\IT3011 - TPSM\\TPSM - Assignment\\cleaned_student_data.csv")

# Re-convert FinalGrade to factor
df$FinalGrade <- as.factor(df$FinalGrade)

# MODEL 1 – Linear Regression (predicting ExamScore)
model1 <- lm(ExamScore ~ StudyHours + Attendance + Discussions +
             Motivation + AssignmentCompletion, data = df)
summary(model1)

# MODEL 2 – Logistic Regression (predicting Pass/Fail)

# Create PassFail variable
# FinalGrade 0 and 1 = Pass (coded as 0)
# FinalGrade 2 and 3 = Fail (coded as 1)
df$PassFail <- ifelse(df$FinalGrade %in% c("2", "3"), 1, 0)

cat("\nPassFail distribution:\n")
table(df$PassFail)

# Build logistic regression model using all 5 reflection proxies
model2 <- glm(PassFail ~ StudyHours + Attendance + Discussions +
              Motivation + AssignmentCompletion,
              data = df, family = "binomial")

summary(model2)

# Predictions and Confusion Matrix
predicted_prob <- predict(model2, type = "response")
predicted_class <- ifelse(predicted_prob > 0.5, 1, 0)

confusion <- table(Predicted = predicted_class, Actual = df$PassFail)
cat("\n--- Confusion Matrix ---\n")
print(confusion)

# Accuracy
accuracy <- sum(diag(confusion)) / sum(confusion) * 100
cat("\nModel Accuracy:", round(accuracy, 2), "%\n")

# Odds Ratios
cat("\n--- Odds Ratios ---\n")
print(round(exp(coef(model2)), 4))
```